

Cross-Lingual Code Retrieval between Indonesian Queries and Python Snippets Using Multilingual E5 Embeddings

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Abstract—This study evaluates cross-lingual code retrieval between Indonesian queries and Python source code and examines whether multilingual embedding models can reduce language mismatch in practical software engineering settings. Existing code retrieval benchmarks assume that users submit queries in English, which limits the applicability of current systems to multilingual developer communities. To address this gap, Indonesian versions of CoSQA queries were generated using two machine translation systems, OPUS MT and Gemma. These translations differ in linguistic fidelity, particularly in the preservation of technical terms and code-related identifiers. Retrieval performance was evaluated using monolingual E5 and Multilingual E5 embedding models. The monolingual E5 model demonstrated a substantial drop when moving from English queries to machine-translated Indonesian queries, while Multilingual E5 maintained significantly stronger performance across all translation conditions. A taxonomy of translation errors was constructed to analyze the impact of literal translations, terminology substitutions, omissions, and hallucinations. The study also includes an embedding drift analysis, which shows that similarity between the English query embedding and the translated version does not reliably predict retrieval accuracy. The determining factor is the semantic alignment between the Indonesian query and the target code snippet. These findings highlight the importance of multilingual training and translation quality for building code retrieval systems that operate effectively in multilingual environments.

Index Terms—cross-lingual retrieval, code search, embeddings, multilingual models, Indonesian NLP

I. INTRODUCTION

Natural language code retrieval plays an important role in software engineering workflows. Developers frequently rely on search engines, documentation platforms and integrated development environments when exploring unfamiliar libraries or locating useful examples. Dense retrieval models have recently become the dominant approach for semantic code search. These models map natural language descriptions and source code into shared embedding spaces and achieve strong performance on established benchmarks [1], [2].

Most existing datasets evaluate code retrieval using only English queries. CoSQA [3] is widely used for Python retrieval but does not include multilingual queries. Indonesian developers often search in Indonesian, which creates a language mismatch that has not been systematically studied. Understanding this

mismatch is essential for practical deployment of code retrieval tools in multilingual environments.

This study addresses two research questions. The first asks whether a multilingual embedding model such as Multilingual E5 [4] provides better Indonesian to Python retrieval performance than the monolingual E5 baseline [5]. The second examines how translation quality affects retrieval outcomes because mistranslating identifiers or technical terminology can substantially alter the intended meaning of a query.

To answer these questions, this work presents the first controlled evaluation of Indonesian to Python retrieval. Indonesian queries are generated using OPUS MT and Gemma, which provide distinct linguistic characteristics. Retrieval performance is measured across six configurations combining monolingual and multilingual embedding models. A taxonomy of translation errors is introduced to analyze how specific error types influence retrieval accuracy and embedding drift.

II. RELATED WORK

Research on code retrieval has advanced rapidly with the development of pretrained text code models and improved benchmarking methodology. CoText [1] demonstrated that multi task pretraining improves semantic code search. UniX-coder [2] introduced a unified cross modal architecture for representing code and natural language.

Parallel work in multilingual retrieval has shown that dense models can align semantically related text across languages. Mr TyDi [6] demonstrated strong multilingual retrieval capabilities across diverse language families. The MTEB benchmark [7] provided a comprehensive evaluation of multilingual embedding quality and revealed several limitations in earlier encoders. Multilingual E5 [4] extends contrastive pretraining to multilingual text pairs, including Indonesian.

Cross linguistic information retrieval research consistently shows that translation quality directly affects retrieval performance. For code retrieval, the sensitivity is higher because identifiers and technical terms often carry functional meaning. The CoIR framework [8] emphasizes the need for controlled evaluation of multilingual retrieval behavior. No prior work has examined Indonesian to code retrieval or analyzed how translation errors influence retrieval accuracy for code specific semantics. This study addresses this gap.

III. METHODOLOGY

A. Dataset

CoSQA [3] contains over twenty thousand English natural language queries paired with Python code snippets. Since the dataset does not include Indonesian queries, this study generates Indonesian versions using two machine translation systems.

B. Query Translation

Two translation systems are used to generate Indonesian queries. OPUS MT tends to produce literal renderings and occasionally alters technical terminology. Gemma 2B produces more fluent Indonesian and more reliably preserves code related identifiers. These differences provide natural variation for assessing the influence of translation quality.

C. Embedding Models

The monolingual E5 base v2 model [5] serves as the English baseline. Multilingual E5 [4] is trained on multilingual aligned corpora and includes Indonesian. Queries follow the required E5 format in which natural language inputs are encoded using the prefix “query:” and code snippets using “passage:”.

D. Retrieval Protocol

Retrieval follows the CoIR framework [8]. Queries are encoded and ranked against code embeddings using cosine similarity. NDCG@10 is used as the evaluation metric.

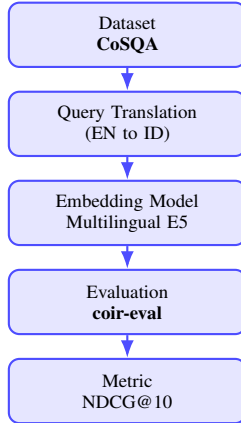


Fig. 1: Pipeline for cross linguistic code retrieval

IV. RESULTS

A. Main Retrieval Performance

Table I presents NDCG@10 scores across six configurations. E5 base v2 performs well on English queries but experiences significant degradation when evaluated on Indonesian translations. Multilingual E5 performs comparably on English queries and substantially better on both Indonesian variants. Gemma translations consistently outperform OPUS translations.

Model Configuration	NDCG@10
E5 base v2, English query	0.332
E5 base v2, Gemma translation	0.135
E5 base v2, OPUS translation	0.077
Multilingual E5, English query	0.315
Multilingual E5, Gemma translation	0.235
Multilingual E5, OPUS translation	0.213

TABLE I: Retrieval performance for English and Indonesian queries

B. Translation Error Categories

Translation errors were annotated into four categories: literal translations, terminology errors, omissions and hallucinations. These categories capture distinct types of semantic distortion. Figure 2 presents the distribution of error types. OPUS produces more literal translations and hallucinations than Gemma. Gemma yields fewer structural errors and generally preserves identifiers more reliably.

C. Translation Quality

Table II highlights clear contrasts between the outputs of Gemma and OPUS, showing that Gemma generally maintains closer lexical fidelity to the original English queries while OPUS tends to introduce greater variation. Gemma more consistently preserves crucial technical identifiers such as *linked list*, *SHA1* and *dictionary*, which helps maintain semantic alignment with Python code snippets where these terms frequently appear verbatim. OPUS, on the other hand, often substitutes these identifiers with Indonesian paraphrases that sound natural in everyday language but ultimately weaken the embedding model’s ability to retrieve the correct code due to reduced lexical and semantic correspondence.

English Query	Indonesian Translations
reverse a linked list	Gemma: membalik linked list OPUS: membalik daftar tertaut
hash a string using SHA1	Gemma: hash string menggunakan SHA1 OPUS: mengacak string menggunakan SHA1
flatten a nested dictionary	Gemma: meratakan dictionary bertingkat OPUS: meratakan kamus bersarang

TABLE II: Comparison of Indonesian translations from Gemma and OPUS

D. Retrieval Impact of Error Types

The impact of each error category is quantified in Table III. Hallucination errors produce the largest average reduction in retrieval performance because they introduce content not present in the original query. Literal translations also reduce

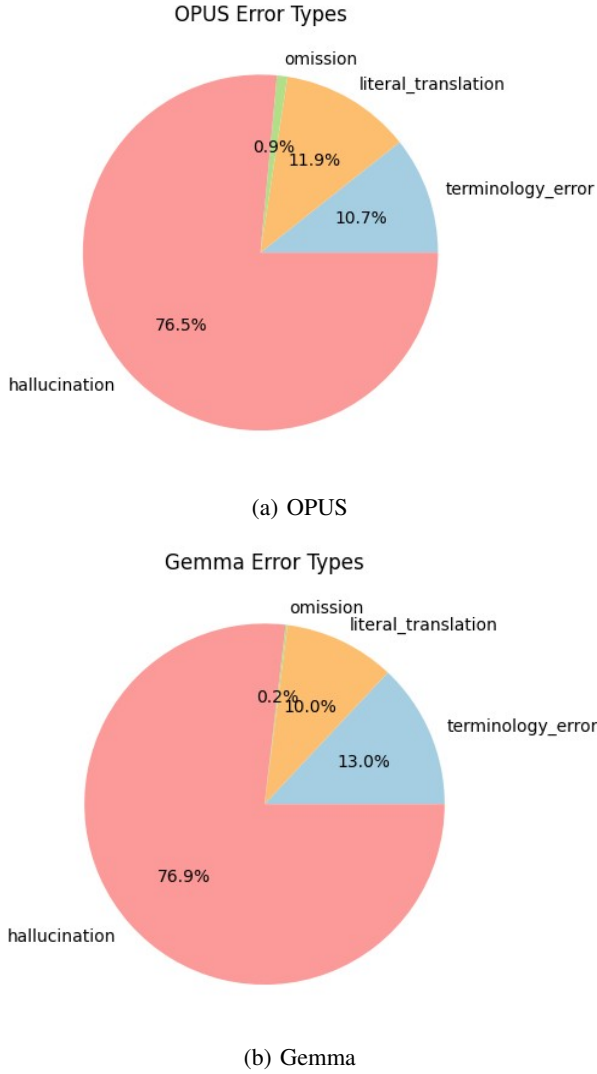


Fig. 2: Distribution of translation error categories

accuracy although some semantic intent is preserved. Terminology errors vary in severity depending on the role of the identifier in the code snippet. Omissions are infrequent but can be highly damaging when key information is removed.

Category	OPUS+E5		OPUS+mE5		Gemma+E5		Gemma+mE5	
	Mean	n	Mean	n	Mean	n	Mean	n
omission	-0.388	6	-0.317	6	-1.000	1	-1.000	1
literal	-0.234	76	-0.112	76	-0.157	62	-0.031	62
hallucination	-0.231	488	-0.101	488	-0.181	479	-0.077	479
terminology	-0.165	68	-0.032	68	-0.175	81	-0.055	81

TABLE III: Mean reduction in NDCG@10 for each translation error category

The patterns in Table III reveal clear differences between translation systems and embedding models. OPUS produces substantially more hallucinations than Gemma, and these errors consistently lead to the largest performance degradation across all configurations. Gemma translations contain more terminology inconsistencies, but the resulting accuracy loss is

relatively small because the overall semantic structure of the query remains intact. Multilingual E5 mitigates the negative effect of every error category, suggesting that it forms a more stable cross-lingual embedding space that is less sensitive to morphological or lexical variation in Indonesian queries. These results indicate that translation errors do not affect all retrieval models equally and that multilingual pretraining provides strong robustness to noisy or imperfect translations.

E. Embedding Drift

Embedding drift measures the cosine distance between the embedding of an original English query and the embedding of its Indonesian translation. This metric reflects how much the translated query deviates from the semantic and lexical structure encoded in the English version. Figures 4 and 5 illustrate these drift patterns for Multilingual E5 and E5 base v2, respectively, broken down by the annotated translation error categories.

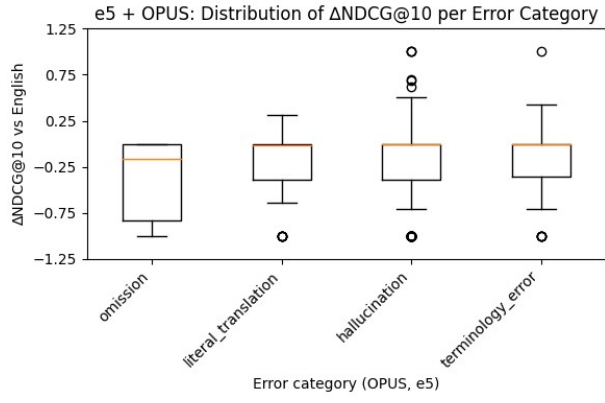
As shown in Figure 4, OPUS translations consistently appear on the lower end of the drift range for Multilingual E5. This occurs because OPUS often applies literal, word-by-word phrasing that keeps the translated text close to the English source in surface structure. Despite this, the figure also shows that OPUS drift values remain low even in categories such as hallucination or terminology errors, indicating that proximity in embedding space does not guarantee semantic fidelity to the original intent.

In contrast, Figure 5 reveals that Gemma translations tend to produce higher drift values under both embedding models. The increased drift reflects Gemma’s more fluent and idiomatic Indonesian phrasing, which naturally diverges from the English input at the lexical and syntactic levels. Importantly, neither figure shows a consistent relationship between drift magnitude and retrieval performance. For instance, Gemma often exhibits larger drift in all error categories, yet retrieval accuracy is consistently higher compared to OPUS for both models.

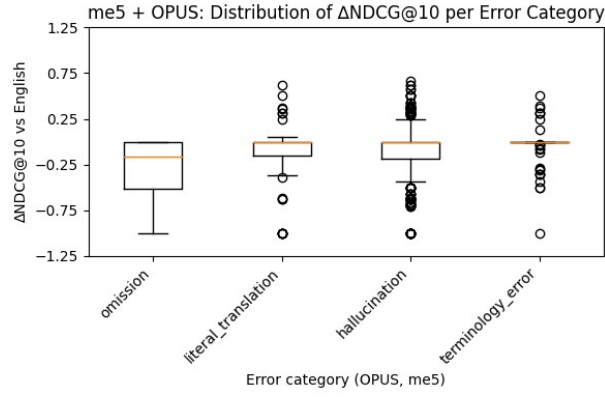
Taken together, the trends in Figures 4 and 5 demonstrate that embedding drift magnitude does not reliably correspond to retrieval quality. A translated query may remain extremely close to the English embedding yet still fail to retrieve the correct snippet if key identifiers or semantic relations are distorted. Conversely, a query may drift farther away while still ranking highly if the translation preserves the technical intent and functional terminology necessary for code matching. These findings reinforce that the determining factor is not similarity to the English embedding, but alignment between the Indonesian query and the code snippet within the shared embedding space.

F. Correlation with Retrieval

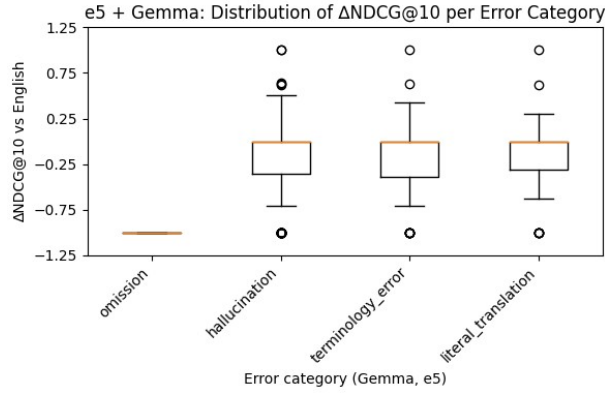
Table IV presents the Pearson correlations between embedding drift and the change in retrieval performance across all model and translation configurations. Across the board, the correlations are extremely close to zero, indicating that the degree of deviation from the English embedding does not meaningfully predict whether retrieval accuracy improves or



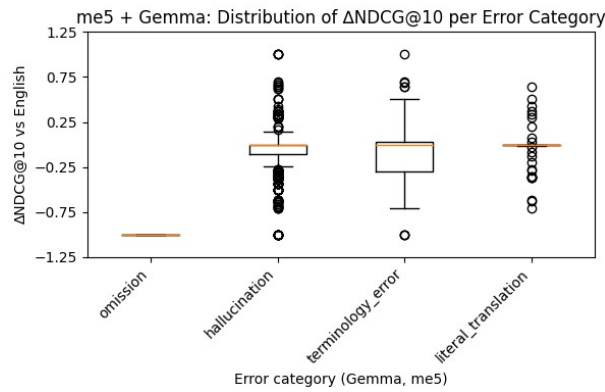
OPUS+E5



OPUS+me5



Gemma+E5



Gemma+me5

Fig. 3: Distribution of error impacts across models and translators

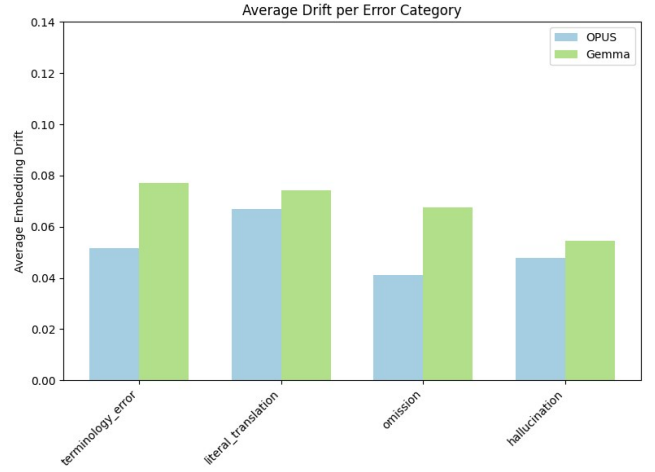


Fig. 4: Embedding drift distribution for Multilingual E5

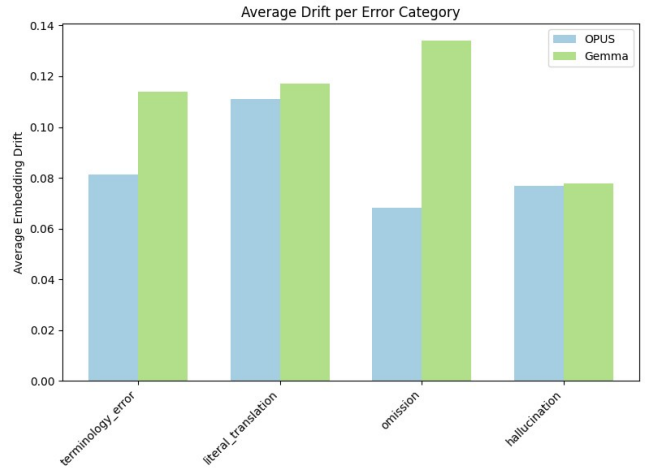


Fig. 5: Embedding drift distribution for E5 base v2

deteriorates. This finding reinforces the observations from the drift analysis: even when a translated query stays close to its English counterpart in embedding space, retrieval quality is not guaranteed.

The lack of correlation suggests that drift alone is an insufficient proxy for semantic preservation. In practice, retrieval performance depends far more on whether the translation captures the functional intent and technical terminology necessary to match the correct Python snippet. Cases in which OPUS translations exhibit low drift yet perform poorly illustrate this disconnect. Conversely, Gemma translations may drift farther from the English embedding while still producing higher-ranking results due to more accurate representation of the underlying task described by the query.

These patterns show that alignment with the target code embedding is the dominant factor governing retrieval success. Embedding drift relative to the English query may reflect linguistic differences between English and Indonesian, but it does not determine how effectively the query aligns with the structure and semantics of the code snippet. As a result,

embedding drift should not be used as a standalone diagnostic for translation quality in cross-lingual code retrieval settings.

Model	Translator	Pearson r	Interpretation
E5 base v2	OPUS	-0.003	No correlation
E5 base v2	Gemma	-0.019	No correlation
Multilingual E5	OPUS	0.019	No correlation
Multilingual E5	Gemma	0.037	No correlation

TABLE IV: Correlation between embedding drift and retrieval change

V. CONCLUSION

This study provides the first systematic evaluation of Indonesian to Python code retrieval. Multilingual E5 performs substantially better than E5 base v2 on Indonesian queries. Translation quality significantly influences retrieval outcomes. Gemma translations consistently outperform OPUS translations due to improved identifier preservation. Embedding drift relative to English does not correlate with retrieval accuracy. Retrieval success depends on semantic alignment between the Indonesian query and the code snippet. Future work may include Indonesian code retrieval datasets based on real developer queries and fine tuning multilingual models.

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